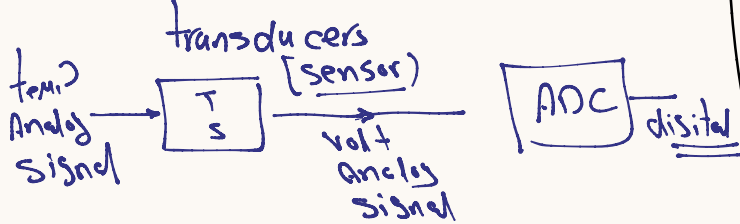
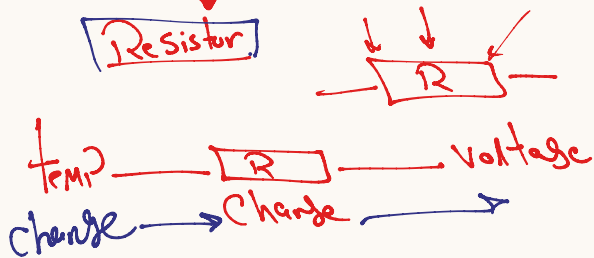
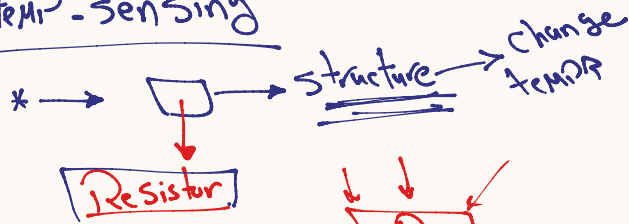


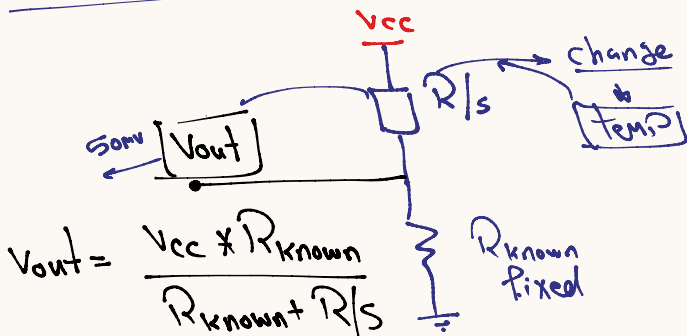
* ADC $\rightarrow$ volt signal only



* temp-sensing



* voltage divider



thermistor $\rightarrow$ Resistor

$\rightarrow$ NTC :: Negative temp Coefficient.

$$R \propto \frac{1}{T}$$

$$\begin{matrix} T \uparrow & \rightarrow & R \downarrow \\ T \downarrow & \rightarrow & R \uparrow \end{matrix}$$

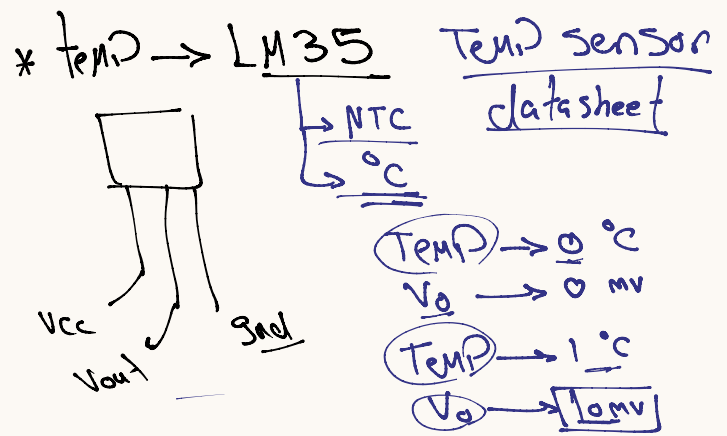
(-40 : 110)

low Price

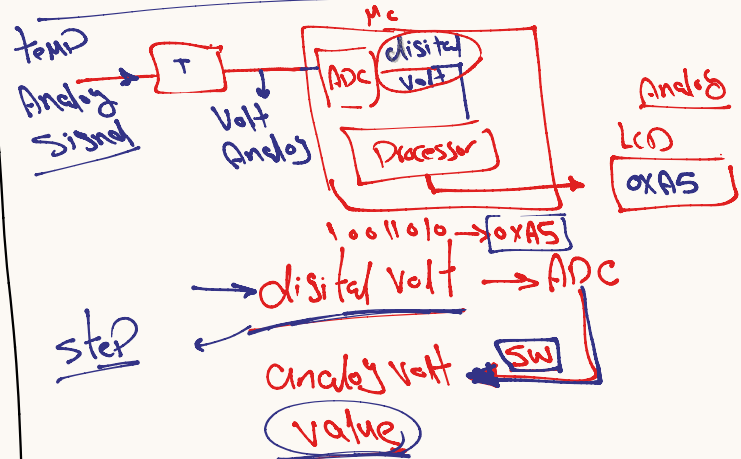
$\rightarrow$ PTC :: Positive temp Coefficient.

$$R \propto T$$

$$\begin{matrix} \text{Temp} \uparrow & R \uparrow \\ \text{Temp} \downarrow & R \downarrow \end{matrix}$$



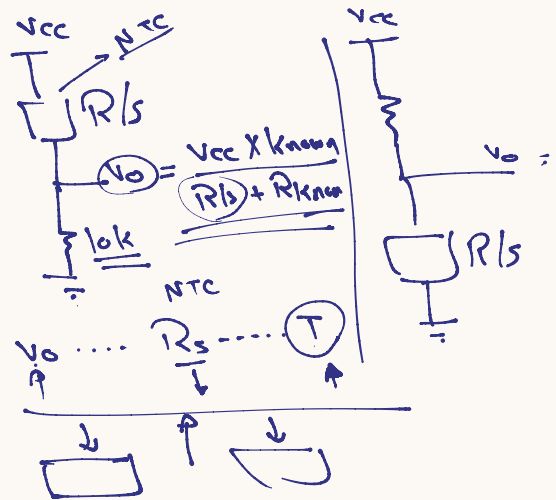
InterFace with ADC



$$\text{digital volt} \approx \text{analog volt}$$

$$\text{Analog} = \text{digital} \times \text{step}$$

voltage divider



* Analog = digital * step

$$\rightarrow \text{ADC-Atmega} \rightarrow \text{step} = \frac{\text{Max volt}}{2^{\text{res}}} = \frac{V_{\text{ref}}}{2^{10}}$$

* Analog = digital * $\frac{V_{\text{ref}}}{2^{10}}$

Analog = (digital * V_{ref}) / 2^{10} ;

assume $\rightarrow V_{\text{ref}} \rightarrow \underline{5V}$

Compiler

uint8_t

Analog = (digital * 5) / 1024

Analog = ((uint32_t) digital * 5UL)

Casting

unsigned Long

* Analog volt = (digital volt * step)

* Coding

#define Vref 5UL $\xrightarrow{5V}$ 5000UL

Analog volt = (uint16_t) digital * Vref / 1024UL ;

uint16_t $\rightarrow$ volt $\rightarrow$ mvolt $\rightarrow$ mvolt

Analog volt $\rightarrow$ ✓ LM35

temp $\rightarrow$ 100u

IC

LM35

Analog

10

$\rightarrow \text{temp} = \frac{\text{Analog volt}}{10} \#$

Must the Analog volt take by millivolt

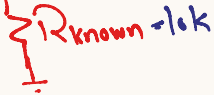
LDR :: Lighte dependent Rrestance

→ NTC

Vcc = 5V



$$V_o = \frac{V_{cc} \times R_{known}}{LDR + R_{known}}$$



light (lux)	LDR (Ω)	V _o →
Deep Dark → 0 lux	> 1MΩ	≈ 0 (0.05V)
→ 10 lux	50kΩ	≈ 0.83V

300 lux: 500

10 klux

> 50 klux

Mapping

Value in Range 1

Range → Represent another Value in another Range

Value → Max

Min ← Value → Max

Range 1: x (Min₁ to Max₁ = 255)

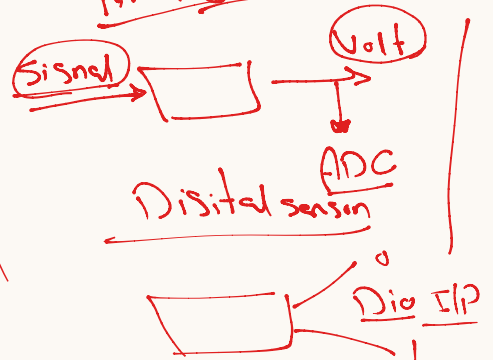
Range 2: y (Min₂ to Max₂ = 8)

③

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$

$$\frac{Max_2 - Min_2}{Max_1 - Min_1} = \frac{y - Min_2}{x - Min_1}$$

Analog Sensor



Ultrasonic → Module

LM35 → Single

LDR → Single

IR → Digital Sensor.

→ 0 (Switch Input)

Rain Sensor

Touch sensor

PIR sensor

Smoke Sensor (MQ)

Gas sensor